Magnetic resonance as Kron's rotating frame

NMR and ESR solved as a three-winding Kron machine, and the bench measurement that discriminates the ring-electron hypothesis. CLAUDE.md §3b; register entry H21

kronspice simulation report · 2026-11-07 · proton at 1 T, T1 = 5 µs, T2 = 2 µs, 400 points per Larmor period

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Summary

Bloch's equations for nuclear and electron magnetic resonance are a rotation of the magnetization about the field, and a rotation of currents at a fixed rate is Kron's frame II(b). This example solves NMR for the proton (and gives the ESR numbers for the electron) with the circuit engine's N machine element: three windings carry the magnetization components as currents, a skew-symmetric speed matrix carries the field, resistors carry the relaxation, and a fourth winding is the pickup coil. It is a companion to *kron_electron* (H21): the resonance frequency is the bench measurement that tells a ring electron from the measured one.

- **Free precession at 42.5766 MHz** against $\gamma B_0/2\pi$ = 42.5775 MHz (-2×10^{-5}, the trapezoidal phase error at 400 points per period).
- **Relaxation exact to the printed digits:** transverse decay 0.01832 at 4T2 (e^{-4} = 0.01832); Mz recovery 0.79810 ($1 - e^{-4T2/T1}$ = 0.79810).
- **Rabi nutation and the π/2 pulse:** period 23.487 µs at B_1 = 1 mT, Mz − cos($\gamma B_1 t$) within 2.4×10^{-4}, zero shaft power; a 5.87 µs pulse leaves M = (0, 1.0000, 0).
- **The continuous-wave line equals Bloch's steady state** to 5 digits at seven detunings.
- **Ledger closes:** source − heat − Δstored = -3×10^{-14} J during free precession.
- **The discriminating measurement for H21:** a ring with the particle's charge and mass (g = 1) would resonate at 13.996 GHz (electron) and 7.623 MHz (proton) at 1 T, against the measured 28.025 GHz and 42.578 MHz.

The construction

Bloch's equations:

$$dM/dt = \gamma \cdot M \times B - (Mx, My, 0)/T2 - (0, 0, Mz - M0)/T1$$

where M is the magnetization, γ the gyromagnetic ratio, B the field, T1 and T2 the longitudinal and transverse relaxation times and M0 the equilibrium magnetization. Kron's frame II(b) (*Equivalent Circuits of Electric Machinery*, 1951, Epilogue eq. 9, p. 257) has constant inductances and a skew-symmetric speed matrix G, so that $L \cdot di/dt = -\omega \cdot G \cdot i$. The mapping:

- Three windings x, y, z carry Mx, My, Mz as their currents (L = 1 nH, so the winding voltages ωL·M are volts).
- $G = L\cdot\gamma\cdot[B]_\times$, the matrix of M × B, and the shaft turns at ω = 1. With $B_{eff} = (B_1, 0, B_0 - \omega_{rf}/\gamma)$ the non-zero entries are $G[x][y] = -L\gamma B_z$, $G[y][x] = L\gamma B_z$, $G[y][z] = -L\gamma B_1$, $G[z][y] = L\gamma B_1$. A skew G has zero torque $i^T \cdot G \cdot i$, so the precession takes no shaft power.
- The NMR rotating frame is Kron's rotating reference frame (Park's transform, CLAUDE.md §3b): at the RF frequency ω_{rf} the RF field B_1 is a constant x-field and the static field becomes $B_0 - \omega_{rf}/\gamma$, so G is constant and resonance is where the effective z-field vanishes; ω_{rf} = 0 is the laboratory frame.
- Relaxation is resistance on the windings: $T2 = L/R_2$ on x and y, $T1 = L/R_1$ on z with a series source $R_1 \cdot M_0$ that restores equilibrium. The heat is on the ledger.
- A fourth, open winding (1 MΩ) coupled to x by M_p = 0.1 nH is the pickup coil; its voltage $M_p \cdot dMx/dt$ is the spectrometer's signal.

Proton run: $\gamma_p = 2.675\,221\,87\times10^8$ rad/s/T, B_0 = 1 T, M_0 = 1 in its own units, T1 = 5 µs and T2 = 2 µs (chosen short so the runs stay small), 400 time points per Larmor period. *BlochSteady* is Bloch's closed-form steady state at a

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detuning, the standard the continuous-wave run is read against.

Results

Proton at 1 T

check	machine	expected
free precession	42.5766 MHz	$\gamma B_0/2\pi$ = 42.5775 MHz (-2×10^{-5}, the trapezoidal phase error $(\omega h)^2/12$ at 400 points per period; 136,252 points over 4T2)
transverse decay at 4T2	0.01832	e^{-4} = 0.01832
Mz recovery at 4T2	0.79810	$1 - e^{-4T2/T1}$ = 0.79810
Rabi nutation, B_1 = 1 mT, relaxation off	period 23.487 μs; Mz − cos($\gamma B_1 t$) within 2.4×10^{-4} over two periods	$2\pi/\gamma B_1$; shaft power 0
π/2 pulse (5.87 μs)	M = (0, 1.0000, 0)	onto the transverse plane
free-induction decay	42.5766 MHz at the pickup, peak 0.02676 V	$M_p \cdot \gamma B_0 \cdot M_0$ = 0.02675 V, decaying with T2
continuous-wave line, B_1 = 10 μT	absorption, dispersion and Mz at seven detunings Δω·T2 = −3 … 3	Bloch's steady state, to 5 digits
ledger (free precession)	source 8.02×10^{-10} J, heat 9.83×10^{-10} J, stored $5.0\times10^{-10} \rightarrow 3.19\times10^{-10}$ J	source − heat − Δstored = -3×10^{-14} J

The continuous-wave line at B_1 = 10 μT

Δω·T2	My (absorption)	Bloch	Mx (dispersion)	Bloch	Mz
−3	0.00054	0.00054	0.00161	0.00161	0.99999
−1	0.00268	0.00268	0.00268	0.00268	0.99996
−0.5	0.00428	0.00428	0.00214	0.00214	0.99994
0	0.00535	0.00535	0.00000	0.00000	0.99993
0.5	0.00428	0.00428	−0.00214	−0.00214	0.99994
1	0.00268	0.00268	−0.00268	−0.00268	0.99996
3	0.00054	0.00054	−0.00161	−0.00161	0.99999

The absorption is the Lorentzian and the dispersion its odd partner, as Bloch's solution gives; at this B_1 the line is unsaturated (Mz stays at 0.9999).

Resonance frequencies at 1 T: measured against the H21 ring

The H21 ring is a ring with the particle's charge and mass circulating together (g = 1, γ = e/2m).

	measured	ring (g = 1)
proton NMR	42.578 MHz (g_p = 5.5857)	7.623 MHz
electron ESR	28.025 GHz (g_e = 2.0023)	13.996 GHz

- A ring electron whose mass circulates with its charge would resonate at half the ESR frequency (its g ≤ 1 is derived in *kron_electron*), and a ring proton at a fifth of the NMR frequency.
- Standard prediction beside it: the Dirac electron with QED, g = 2.002 319 304 36, and the measured proton g.

Animation

viz/nmr.gif (100 frames) shows the magnetization vector in the rotating frame (left) through the π/2 pulse and the decay that follows, with the pickup coil's signal drawn as it arrives (right). The frames are the solver's time points, the decay run converted to the rotating frame by Park's rotation. It is not reproduced here because a single frame does not show the motion; open the GIF to see it.

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Against existing measurements

The resonance frequencies are measured inputs, so the match is exact by construction: the proton's 42.577 MHz/T is the CODATA value of $\gamma_p/2\pi$, and the electron's 28.025 GHz/T follows from the measured g-factor. What the machine adds is the dynamics, and those are Bloch's 1946 equations, whose predictions (precession, T1 and T2 relaxation, the Lorentzian absorption and dispersion lines, saturation) were confirmed in the first NMR experiments of Purcell, Torrey and Pound and of Bloch, Hansen and Packard (both 1946) and in every spectrometer since; ESR was first observed by Zavoisky in 1945. The scaled relaxation times and pickup units are not measurements.

Validation

test	check	held to
TestFreePrecession	frequency γB_0; $\lvert Mxy\rvert = e^{-2}$ and $Mz = 1 - e^{-2T2/T1}$ at 2T2; ledger closed	1e-4; mismatch 1e-3 of heat
TestRabi	$Mz = \cos(\gamma B_1 t)$ at resonance; π/2 pulse gives (0, 1, 0)	1e-3
TestPickup	pickup peak $M_p \cdot \gamma B_0 \cdot M_0$ at frequency γB_0	1e-3, 1e-4
TestCWLine	absorption, dispersion and Mz at $\Delta\omega \cdot T2 = -2, -0.5, 0, 0.5, 2$ against BlochSteady	2e-5
TestExport	free precession (.tran 50p 8u uic) and the CW run at resonance (.tran 40n 60u uic) exported with the N element	—

Limits

- T1 and T2 are chosen short (microseconds) so the runs stay small; they are not measured values.
- The resonance frequencies use measured g-factors as inputs; the model checks the dynamics, not the magnetic moments.
- Free precession carries a trapezoidal phase error of -2×10^{-5} at 400 points per period.

Status of H21 (context)

H21, the ring electron as a Kron machine, is *bounded* in the register: no ring with the electron's charge-to-mass ratio reaches g = 2. This example supplies the bench measurement that discriminates it: ESR at 28.0 GHz/T measured against 14.0 GHz/T for a ring electron, and NMR at 42.6 MHz/T against 7.6 MHz/T for a ring proton.

Reproducing the results

```
go run ./examples/kron_nmr -gif viz/nmr.gif
go test ./examples/kron_nmr
KRONSPICE_EXPORT=1 go test ./examples/kron_nmr -run TestExport   # netlists, KiCad, results sheets
```

Sources: F. Bloch, Phys. Rev. 70, 460 (1946); F. Bloch, W. W. Hansen, M. Packard, Phys. Rev. 69, 127 (1946); E. M. Purcell, H. C. Torrey, R. V. Pound, Phys. Rev. 69, 37 (1946); E. Zavoisky, J. Phys. USSR 9, 211 (1945); G. Kron, Equivalent Circuits of Electric Machinery, 1951, Epilogue eq. 9; CODATA 2018 (Tiesinga et al., Rev. Mod. Phys. 93, 025010, 2021); D. Hanneke et al., PRL 100, 120801 (2008); X. Fan et al., PRL 130, 071801 (2023).

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